School Worksheet

Class 12 Mathematics, Chapter 10: Vector Algebra

Maximum marks: **47** **20** questions

Name _____ Date _____ Score _____ / 47

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

The unit vector perpendicular to the vectors $\hat{i} - \hat{j}$ and $\hat{i} + \hat{j}$ is:

(A) $\hat{k}$ **(B)** $-\hat{k} + \hat{j}$ **(C)** $\frac{\hat{i} - \hat{j}}{\sqrt{2}}$ **(D)** $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$

2 MCQ — 1 mark

If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$, $|\vec{a}| = \sqrt{37}$, $|\vec{b}| = 3$ and $|\vec{c}| = 4$, then the angle between $\vec{b}$ and $\vec{c}$ is:

(A) $\frac{\pi}{6}$ **(B)** $\frac{\pi}{4}$ **(C)** $\frac{\pi}{3}$ **(D)** $\frac{\pi}{2}$

3 MCQ — 1 mark

The value of λ for which $2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\hat{i} - 2\hat{j} + 3\hat{k}$ are perpendicular is:

(A) $\frac{5}{2}$ **(B)** $-\frac{5}{2}$ **(C)** 2 **(D)** $\frac{3}{2}$

4 MCQ — 1 mark

If $|\vec{a}| = 2$, $|\vec{b}| = 5$ and $\vec{a} \cdot \vec{b} = 8$, then $|\vec{a} \times \vec{b}|$ is:

(A) 2 **(B)** 4 **(C)** 6 **(D)** 8

5 MCQ — 1 mark

The projection of $\vec{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$ is:

(A) $\frac{10}{\sqrt{17}}$ **(B)** $\frac{5\sqrt{6}}{3}$ **(C)** $\frac{10}{6}$ **(D)** $\sqrt{6}$

6 Assertion-Reason — 1 mark

Assertion (A): The vectors $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = 2\hat{i} + 4\hat{j} + 6\hat{k}$ are collinear. **Reason (R):** Two vectors $\vec{a}$ and $\vec{b}$ are collinear if $\vec{b} = \lambda\vec{a}$ for some scalar λ.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Q No.	Question	Marks
7	Short answer Find the area of the parallelogram whose adjacent sides are $\vec{a} = \hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$.	2 marks

Level 2

16 marks

8	Short answer Find a vector of magnitude 21 units in the direction opposite to that of $\overrightarrow{AB}$, where A and B are the points $A(2, 1, 3)$ and $B(8, -1, 0)$ respectively.	2 marks
9	Short answer Find a unit vector perpendicular to both $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.	2 marks
10	Short answer Two friends flying kites from different locations find the strings of their kites crossing each other. The strings can be represented by the vectors $\vec{a} = 3\hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 2\hat{i} - 2\hat{j} + 4\hat{k}$. Determine the angle formed between the kite strings. (Assume there is no slack in the strings.)	2 marks
11	Short answer Find the area of the triangle with vertices $A(1, 1, 2)$, $B(2, 3, 5)$ and $C(1, 5, 5)$.	3 marks
12	Short answer Find the projection of the vector $(\vec{b} + \vec{c})$ on the vector $\vec{a}$, where $\vec{a} = 2\hat{i} + 2\hat{j} + \hat{k}$, $\vec{b} = \hat{i} + 3\hat{j} + \hat{k}$ and $\vec{c} = \hat{i} + \hat{k}$.	3 marks
13	Short answer If $\vec{a} = 2\hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j}$, find λ such that $\vec{a} + \lambda\vec{b}$ is perpendicular to $\vec{c}$.	2 marks
14	Short answer Find the position vector of the point R which divides the line joining $P(2\vec{a} + \vec{b})$ and $Q(\vec{a} - 3\vec{b})$ externally in the ratio $1 : 2$.	2 marks

Level 3

23 marks

15	Short answer If $\vec{a}$, $\vec{b}$, $\vec{c}$ are unit vectors such that $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c} = 0$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{6}$, prove that $\vec{a} = \pm 2(\vec{b} \times \vec{c})$.	3 marks
16	Long answer Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. Find a vector $\vec{d}$ which is perpendicular to both $\vec{a}$ and $\vec{b}$, and $\vec{c} \cdot \vec{d} = 15$.	5 marks
17	Short answer Find a vector of magnitude 5 units perpendicular to each of $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$, where $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$.	3 marks

Q No.	Question	Marks
18	Short answer Using vectors, find the area of the triangle ABC with vertices $A(1, 2, 3)$, $B(2, -1, 4)$ and $C(4, 5, -1)$.	3 marks
19	Long answer Show that the points with position vectors $2\hat{i} - \hat{j} + \hat{k}$, $\hat{i} - 3\hat{j} - 5\hat{k}$ and $3\hat{i} - 4\hat{j} - 4\hat{k}$ are the vertices of a right-angled triangle. Also find the remaining angles' cosines.	5 marks
20	Case study **Case Study:** A toy drone starts from the origin O. It first moves to point A with $\overrightarrow{OA} = 2\hat{i} + 3\hat{j} + \hat{k}$ and is then commanded to point B with $\overrightarrow{OB} = 4\hat{i} + \hat{j} + 3\hat{k}$ (units in metres). **(i)** Find the displacement vector $\overrightarrow{AB}$. — 1 mark **(ii)** Find the distance travelled from A to B and the unit vector along $\overrightarrow{AB}$. — 1 mark **(iii)** Find the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$. — 2 marks	4 marks

Answer key

Chapter 10: Vector Algebra, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(A) $\hat{k}$
2	(C) $\frac{\pi}{3}$
3	(A) $\frac{5}{2}$
4	(C) 6
5	(B) $\frac{5\sqrt{6}}{3}$
6	(A) Both A and R are true and R is the correct explanation of A
7	Area $= 2\sqrt{2}$ sq units

Level 2

8	$-18\hat{i} + 6\hat{j} + 9\hat{k}$
9	$\pm\frac{1}{\sqrt{6}}(\hat{i} - 2\hat{j} + \hat{k})$
10	$\theta = \cos^{-1}\left(\frac{\sqrt{21}}{7}\right)$
11	Area $= \frac{\sqrt{61}}{2}$ sq units
12	Projection $= 4$
13	$\lambda = 8$
14	$\overrightarrow{OR} = 3\vec{a} + 5\vec{b}$

Level 3

15	$\vec{a} = \pm 2(\vec{b} \times \vec{c})$, proved
16	$\vec{d} = \frac{1}{3}(160\hat{i} - 5\hat{j} - 70\hat{k})$
17	$\pm\frac{5}{\sqrt{6}}(-\hat{i} + 2\hat{j} - \hat{k})$, i.e. $\pm\frac{5\sqrt{6}}{6}(-\hat{i} + 2\hat{j} - \hat{k})$
18	Area $= \frac{\sqrt{274}}{2}$ sq units
19	Right angle at C; $\cos A = \sqrt{\frac{35}{41}}$, $\cos B = \sqrt{\frac{6}{41}}$
20	(i) $2\hat{i} - 2\hat{j} + 2\hat{k}$; (ii) $2\sqrt{3}$ m, $\frac{1}{\sqrt{3}}(\hat{i} - \hat{j} + \hat{k})$; (iii) $\cos^{-1}\sqrt{\frac{7}{13}}$